

CareConnect

Software Test Plan

University of Maryland Global Campus

Software Engineering 670 9040 Software Engineering Capstone

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Revision History

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1. Executive Summary

This Software Test Plan defines the testing strategy, processes, environments, responsibilities, and planned validation activities for Team D's assigned CareConnect enhancements. Team D is responsible for Electronic Visit Verification (EVV) enhancements, AI-assisted invoicing migration, QR code and invite-link onboarding workflows, wearable device integrations, home care document digitization, and assessment workflows.

Milestone 2 focuses on planning and design validation rather than execution. Therefore, this document establishes the testing framework, test environments, coverage strategy, acceptance criteria, and integration considerations that will be used during Milestone 3 implementation and Milestone 4 validation activities. The plan also identifies external dependencies, risks, assumptions, and testing approaches for integrations that may not yet be available for production use.

2. Introduction

2.1 Purpose

The purpose of this Software Test Plan is to provide a structured and repeatable approach for validating Team D's assigned CareConnect functionality. This document defines what will be tested, how testing will be performed, who is responsible for testing activities, and how results, defects, and acceptance decisions will be tracked throughout the capstone project lifecycle.

2.2 Scope

This test plan applies to Team D's assigned CareConnect workstreams and associated technical deliverables.

In Scope

- Electronic Visit Verification (EVV) enhancements
- AI-assisted invoicing workflow migration to AWS Bedrock and Claude
- QR code and invite-link onboarding workflows
- Wearable device integrations (Fitbit, Apple HealthKit, Google Health Connect)
- Home care document digitization workflows
- Screening and comprehensive assessment workflows
- Associated APIs, database interactions, user interfaces, and integration points

Out of Scope

- Production EVV aggregator integrations
- Enterprise infrastructure testing
- EHR/FHIR integrations
- Features owned by other CareConnect teams
- Production healthcare billing systems
- Third-party systems without available testing environments

2.3 Intended Audience

Audience	Reason for Using This Document
Team D members	Understand assigned testing responsibilities and test documentation expectations.
Team lead	Review coverage, risks, blockers, and integration needs.
Other SWEN 670 teams	Identify cross-team dependencies, shared test data, and integration points.
SWEN 661 collaborators, if applicable	Understand test expectations for UI or feature work that supports CareConnect.
Instructor / client stakeholders	Evaluate whether planned testing supports milestone goals and final delivery readiness.
Future Development Teams	Understand testing processes and reusable test assets.

2.4 References

Reference	Purpose
Project Plan	Defines milestone deliverables, roles, schedule, risks, and planned work.
Software Requirements Specification (SRS)	Defines functional and nonfunctional requirements to be tested.
Technical Design Document (TDD)	Defines the design that this test plan must validate.
CareConnect Coding Readiness Report	Identifies implementations readiness and dependencies
GitHub Repository and Issue Tracking	Tracks development progress, defects, and blockers
AWS Bedrock Documentation	Supports AI-assisted invoicing design and testing

2.5 Definitions, Acronyms, and Abbreviations

Term	Definition
STP	Software Test Plan.
TDD	Technical Design Document.
SRS	Software Requirements Specification.
Test Case	A documented condition, action, input, and expected result used to verify system behavior.
Defect	A confirmed issue where actual behavior does not match expected behavior.
Regression Test	A test repeated after changes to confirm that existing functionality still works.
Acceptance Criteria	Conditions that must be met before a feature, module, or milestone deliverable is accepted.
EVV	Electronic Visit Verification
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Service
Bedrock	AWS managed AI platform
OCR	Optical Character Recognition
PHI	Protected Health Information
RBAC	Role-Based Access Control
QR Code	Quick Response Code used for onboarding workflows

3. Test Scope

3.1 In Scope

The following features, APIs, services, screens, and cross-cutting concerns are included in Team D's test scope for Milestone 2 planning and Milestone 3 execution.

Item	Description	Reason for Testing	Priority
EVV Workflow (EvvService)	Check-in/check-out lifecycle, geofence validation, pre-export field validation, ARRIVE/DEPART audit trail	CMS-mandated EVV compliance; correction and audit trail directly address prior cohort gaps	High
EVV Aggregator Dashboard	Aggregator dependency tracking widget for 10+ aggregators; status display (PENDING, SUBMITTED, FAILED)	REQ-1401 requires caregiver/admin visibility of aggregator submission status	High
Invoice Extraction (InvoiceService + BedrockAdapter)	Invoice field extraction via MockBedrockAdapter; reviewer continuity; fallback behavior on mock error	Migration from DeepSeek to Bedrock is high-risk; correctness and fallback must be confirmed	High
QR Code/Link Invite System (InviteService)	Token generation (HMAC-SHA256, 72hr TTL), QR code rendering, accept/decline flow, token expiry/replay prevention	Security-sensitive feature; token integrity and RBAC enforcement must be verified	High
Wearable Integration (WearableService)	Health Connect sync, metric persistence, threshold evaluation, FCM alert dispatch, consent revocation	New feature with external integration (FCM); PHI handling and consent revocation are compliance concerns	High
Document Digitization	Form upload acknowledgement (UPLOADED state); design-phase coverage of W-4, I-9, Direct Deposit workflow	Milestone 2 is design phase only; partial coverage confirms upload endpoint; full implementation deferred to milestone 3	Medium
Screening Intake Form (ScreeningService)	Form creation, draft/save/submit lifecycle, required-field validation, immutability after submission	New feature; submission immutability and field validation protect clinical record integrity	High
Comprehensive Assessment (AssessmentService)	5-section assessment lifecycle, section-level draft save, all-sections required validation, amendment workflow	Complex multi-part form; amendment workflow and immutability are patient safety concerns.	High
Security/RBAC	Role enforcement on all Team D endpoints (ADMIN, SUPERVISOR, CAREGIVER, PATIENT); PHI exclusion from logs; invite token security	HIPAA compliance; unauthorized access and PHI leakage must be detected and rejected	High
NFR/Performance	EVV response time P95 <1s; token validation <300 ms; batch wearable sync <3s; TLS 1.3 enforcement	Performance and transport security NFRs defined in SRS; baseline required before Milestone 3 load testing	Medium
Regression	Prior cohort EVV, Invoice, and Invite workflows unaffected by Team D changes	Prevent regressions in production-critical workflows; baseline must survive Team D enhancements	High
Cross-Team Integrations	RBAC constants from Team A; CI/CD pipeline; Bedrock	Integration points affect multiple teams; misalignment	High

Item	Description	Reason for Testing	Priority
	provisioning; FCM payload PHI check; S3 SSE-KMS encryption	detected early reduces Milestone 3 risk	

3.2 Out of Scope

Identify items that will not be tested by this team unless assigned later. This prevents confusion and helps other teams understand ownership boundaries.

Out-of-Scope Item	Reason	Potential Owner / Dependency
Live AWS Bedrock invocation	Pending Anthropic account approval; not provisioned in Milestone 2.	Team A (infrastructure provisioning)
HealthKit (iOS) wearable integration	iOS-specific SDK; deferred to Milestone 3	Milestone 3 implementation
Fitbit and third-party wearable APIs	Out of SRS scope; future milestone only	Future Milestone
Document OCR/text extraction	Document digitization is in design-phase only in Milestone 2; OCR deferred to Milestone 3	Milestone 3 Implementation
WCAG 2.1 AA formal accessibility audit	Delegated to Team B per project plan	Team B
Multilingual/localization testing	Owned by Team C per project plan	Team C
Team A infrastructure (RBAC, provisioning, video calls, telemetry, analytics)	Outside Team D feature scope	Team A
Team E features (USPS Informed Delivery, STML/AI agent, call/visit summaries)	Outside Team D feature scope	Team E
EHR/FHIR integration	Future Milestone; not in SRS scope for Milestone 2	Future Milestone
Electronic signature workflows	Not assigned to Team D	Not assigned
RDS backup/failover (RPO/RTO NFR-05)	Owned by Team A; requires infrastructure access	Team A
Production AWS environment testing	All Milestone 2 and Milestone 3 testing uses mock adapters and development environments	Team A Production Deployment

3.3 Testing Objectives

- Verify that all eight Team D feature areas behave as specified in the SRS and designed in the TDD.
- Validate that the technical design can be tested, integrated, and demonstrated with mock adapters in the absence of live external services.
- Identify defects, blockers, and open questions early so they can be resolved before Milestone 3 execution.
- Support consistent class-wide documentation and milestone evidence by mapping all 47 requirements and FNRS to specific test cases.
- Prepare reusable, executable test assets for Milestone 3 implementation and Milestone 4 final test reporting.

4. Test Items and Features

4.1 Test Item Inventory

Test Item ID	Test Item	Type	Description	Owner
TI-001	EvvService.checkIn()	Functional/API	GPS geofence validation, visit status transition SCHEDULED → IN_PROGRESS, audit entry creation	Cristobal Garcia
TI-002	EvvService.checkOut()	Functional/API	Check-out field validation, DEPART audit entry, visit status transition IN_PROGRESS → COMPLETED	Cristobal Garcia
TI-003	EvvService.correctVisit()	Functional/API	Visit correction workflow; EvvVisitCorrection entity with before/after JSONB snapshot; RBAC enforcement	Cristobal Garcia
TI-004	EvvService.validatePreExport()	Functional/API	Pre-export validation of all required fields before aggregator submission	Cristobal Garcia
TI-005	EvvVisitAuditEntry()	Functional	Audit trail record for each visit state transition; confirms eventType, gpsHash, actorId, timestamp	Cristobal Garcia
TI-006	AggregatorDependency Widget	UI/Integration	Admin dashboard widget displaying 10+ aggregator statuses; PENDING/SUBMITTED/FAILED indicators	Cristobal Garcia
TI-007	InvoiceService (Bedrock extraction)	Functional/Integration	Invoice field extraction via MockBedrockAdapter; structured JSON output mapping; reviewer continuity	Cristobal Garcia
TI-008	BedrockAdapter (mock fallback)	Integration	Fallback behavior when MockBedrockAdapter returns error; InvoiceService graceful degradation	Cristobal Garcia
TI-009	InvoiceService (PDF generation)	Functional/API	POST /api/v1/invoices → PDF generated, stored via S3Adapter, invoice record created	Cristobal Garcia
TI-010	InviteService.generateToken()	Functional/Security	HMAC-SHA256 token generation, 72hr TTL enforcement, QR code rendering, copyable URL	Nhial Mayar
TI-011	InviteService.validateToken()	Functional/Security	Token signature verification, expiry enforcement, replay prevention (one-time accept), role assignment	Nhial Mayar

Test Item ID	Test Item	Type	Description	Owner
TI-012	InviteRecord lifecycle	Functional	PENDING → ACCEPTED/DECLINED status transitions; RBAC enforcement on invite creation (ADMINS/SUPERVISOR only)	Nhial Mayar
TI-013	WearableService.syncMetrics()	Functional/Integration	Health Connect batch payload ingestion, metric persistence, threshold evaluation, FCM alert dispatch	Davon Davis
TI-014	WearableService (consent revoke)	Functional/Security	Consent revocation stops future syncs; CONSENT_ALREADY_REVOKED enforcement	Davon Davis
TI-015	MockFCM notification payload	Integration/Security	FCM notification contains metric type, value patient identifier; no full PHI in payload	Davon Davis
TI-016	DocumentService.uploadDocument()	Functional/API	File upload acknowledgement; record created in UPLOADED state	Bernabe Garcia
TI-017	ScreeningService.createRecord()	Functional/API	Screening form creation, DRAFT → SUBMITTED lifecycle, required field validation, immutability	Angel Guanlao
TI-018	AssessmentService (comprehensive)	Functional/API	5-section assessment create/draft/save/submit lifecycle, amendment workflow, immutability after submission	Angel Guanlao
TI-019	RBAC enforcement (all endpoints)	Security	Role-based access control enforced on all Team D REST endpoint; 403 for unauthorized roles	Bernabe Garcia
TI-020	PHI log exclusion	Security/Privacy	Confirm raw GPS, patient name, DOB absent from CloudWatch/application logs; gpsHash used instead	Bernabe Garcia
TI-021	EVV response time P95	Performance	EVV check-in/check-out response time <1 second at P95	Cristobal Garcia
TI-022	Invite token validation latency	Performance	Token validation response time <300 ms P95	Nhial Mayar
TI-023	Wearable batch sync latency	Performance	Batch of 100 metrics processed in <3 seconds	Davon Davis
TI-024	TLS 1.3 enforcement	Security/NFR	Confirm all Team D API traffic uses TLS 1.3; HTTP and TLS 1.2 connections rejected at load balancer	Bernabe Garcia
TI-025	S3Adapter SSE-KMS encryption	Security/Integration	Invoice attachments stored with	Cristobal Garcia

Test Item ID	Test Item	Type	Description	Owner
			serviceSideEncryption=aws:kms; no AES256 or omission	
TI-026	Flutter screens (13 Team D screens)	UI	Smoke-level UI verification for all 13 Team D Flutter screens; navigation, form rendering, error display	Angel Guanlao
TI-027	Regression: prior cohort EVV/Invoice/Invite	Regression	Existing check-in, invoice review, and invite acceptance flows unaffected by Team D changes	Cristobal Garcia/Nhial Mayar

4.2 Features to Be Tested

Feature ID	Feature / Workflow	Requirement Reference	Expected User or System Outcome
FT-001	EVV Workflow Gap Closure	REQ-1401, REQ-1402, REQ-1403, REQ-1404	Caregiver check-in/check-out validated against GPS geofence; corrections tracked with before/after snapshot; audit trail written for each transition; export blocked on missing fields.
FT-002	Aggregator Dependency Tracking	REQ-1401	Admin dashboard displays real-time submission status for 10+ EVV aggregators; PENDING, SUBMITTED, and FAILED states visible; last-updated timestamp shown.
FT-003	Invoice Extraction Migration	REQ-1501-REQ-1507	Invoice fields extracted via MockBedrockAdapter; structured output formatted for reviewer continuity; fallback returns graceful error when adapter fails; PDF generated and stored via S3Adapter
FT-004	QR Code/Link Invite System	REQ-033-REQ-038	Invite tokens generated with HMAC-SHA256 and 72hr TTL; QR code and URL both deliver valid token; token accepted or declined once; expired and replayed tokens rejected with correct error codes.
FT-005	Wearable Device Integration	REQ-091-REQ-095	Health Connect metrics synced and persisted; threshold rules evaluated; FCM alert dispatched to configured caregiver when threshold breached; 60-minute cooldown prevents alert flooding; consent revocation blocks future syncs
FT-006	Home Care Document Digitization	REQ-1601-REQ-1607	Document upload endpoint returns UPLOADED state; W-4, I-9, and Direct Deposit digitization workflows

Feature ID	Feature / Workflow	Requirement Reference	Expected User or System Outcome
			designed and documented; OCR and data extraction deferred to Milestone 3.
FT-007	Screening Intake Form	REQ-1701-REQ-1706	Screening record created by CAREGIVER; DRAFT saved on field change; submitted record is immutable; required fields validated server-side; 409 returned on modification attempt after submission.
FT-008	Comprehensive Assessment	REQ-1801-REQ-1809	5-section assessment created; section-level drafts saved; all 5 sections required for submission; submitted record is immutable; amendment creates a new linked record without modifying the original.

4.3 Features Not to Be Tested

Feature / Area	Reason Not Tested	Future Handling
Live AWS Bedrock invocation	Bedrock account not provisioned; API keys not available in Milestone 2	Milestone 3 - Team A provisions Bedrock access; MockBedrockAdapter replaced with live adapter
HealthKit (iOS) wearable SDK	iOS-specific integration; Android Health Connect is Milestone 2 scope	Milestone 3 - implements iOS Health Connect path
Document OCR/text extraction	Design phase only in Milestone 2; extraction logic not yet implemented	Milestone 3 - document digitization full implementation
WCAG 2.1 AA formal audit	Delegated to Team B per project assignment	Team B - post Milestone 2 accessibility certification
Multilingual/localization	Owned by Team C; not in Team D scope	Team C
RDS backup/failover (NFR-05)	Infrastructure-level; requires Team A AWS environment access	Team A - Infrastructure testing
Electronic signature workflows	Not assigned to Team D	Not assigned
Fitbit and third-party wearables	Outside SRS scope for current milestones	Future milestone

5. Testing Strategy

5.1 Overall Approach

Risk-based, shift-left testing strategy: The CareConnect testing strategy operates concurrently with our Agile-Scrum development model, enforcing testing activities within the weekly sprints rather than at the end of the lifecycle. By "shifting left," we execute Static Application Security Testing (SAST) and Software Composition Analysis (SCA) automatically on every pull request, preventing vulnerable dependencies from entering the main branch. This approach aligns with our four-milestone timeline (M1 05/30, M2 06/13, M3 07/25, M4 08/04), distributing the QA workload evenly and preventing integration bottlenecks.

Healthcare-first mindset: Operating within a highly regulated domain, every single test case is designed with security and compliance as primary acceptance criteria. Our test artifacts explicitly map to the HIPAA Privacy Rule, Security Rule, and Breach Notification Rule. Furthermore, we test for strict adherence to the HITECH guidelines and the 21st Century Cures Act §12006 (mandating electronic verification of caregiver, patient, location, service type, and time). Test execution provides documented evidence of compliance.

V-Model traceability: We employ a strict Requirements Traceability Matrix (RTM) linking the Software Requirements Specification (SRS) down to individual test executions. Every SRS requirement (from REQ-033 through REQ-5.11.x) maps to at least one functional test case. Where applicable, these functional flows are augmented by targeted security, performance, or resilience tests to ensure non-functional constraints (like TLS 1.3 enforcement and p95 latency targets) are met.

Test pyramid & Compliance pyramid: We balance our test automation efforts to achieve approximately 70% Unit coverage, 20% Integration/API coverage, and 10% End-to-End (E2E) and exploratory coverage. Running parallel to this is our "compliance pyramid," which mandates low-level ORM checks for data encryption, mid-level API tests for RBAC enforcement across Caregiver/Patient/Admin roles, and high-level E2E tests validating the immutable audit trail capture.

Reference Figure 2 Technical Design Document: Testing efforts are structurally aligned with the data domains established in Figure 2 of the Technical Design Document. We segment test suites across the three core swim lanes: EVV (validating, etc.), AI Invoicing (validating visit, offline sync queue, etc), AI Invoicing (validating invoice, extraction result), and Care Ops/Identity/Audit (validating user, audit event, etc.). Specific append-only validation logic strictly targets the audit event and model prompt log tables.

5.1.1 Testing Philosophy

1. **Shift-Left Security:** Security is not an afterthought. Every Pull Request triggers automated SAST/SCA checks in the CI/CD pipeline managed by Team A. Code cannot merge if critical vulnerabilities are detected.

2. **Defense-in-Depth Testing:** We validate controls at multiple layers. We test input validation at the Flutter UI, JWT signature verification at the API Gateway, RBAC enforcement via Spring Security at the controller layer, and constraint validation at the PostgreSQL ORM layer.
3. **Compliance-as-Code:** HIPAA controls are not just documented; they are asserted via automated tests. We programmatically verify that AES-256 is used for database encryption and TLS 1.3 is negotiated for API traffic.
4. **Append-Only Verification:** We actively attempt to execute unauthorized `UPDATE` and `DELETE` SQL commands against the audit event and model prompt log tables to prove that database-level triggers correctly reject history tampering.
5. **Human-in-the-Loop Validation:** To satisfy REQ-1507, automated tests simulate attempts to force an AI-extracted invoice into an "accepted" state without human review, verifying that the system rejects the operation regardless of AI confidence scores.
6. **Synthetic-Only Data:** Real Protected Health Information (PHI) is absolutely prohibited in all non-production environments. All test data generation complies with the HIPAA Safe Harbor de-identification standard (§164.514).

5.1.1 Alignment to Project Plan Milestones

Milestone	Date	QA / Testing Deliverables
M1	05/30/2026	Testing strategy drafted; high-level requirements (REQ IDs) ingested into test management; Tooling access requested.
M2	06/13/2026	Software Test Plan v1.0 finalized (this document); Initial test cases written; Mock data generators built; BAT suite configured.
M3	07/25/2026	Full automated test execution pipeline active; Initial performance baselines established via k6; Monthly penetration test executed.
M4	08/04/2026	Test Report completed; 100% RTM coverage achieved; Zero Sev-1/Sev-2 defects remaining; Formal QA sign-off granted.

5.1.2 Testing Scope by Module

Onboarding & Identity: REQ-033 to REQ-038 (Invite tokens, QR delivery, RBAC role assignment).

Wearables Integration: REQ-091 to REQ-095 (OAuth flows, data ingestion, threshold alerts, stale sync tracking).

EVV Operations: REQ-1401 to REQ-1404 (Offline check-ins, geofence rules, aggregator EDI generation).

AI Invoicing: REQ-1501 to REQ-1507 (Bedrock/Claude API mocks, confidence scores, human approval gating).

Document Digitization: REQ-1601 to REQ-1607 (Form validation, file type constraints, administrative review).

Screening & Assessment: REQ-1701 to REQ-1706 and REQ-1801 to REQ-1809 (Clinical intake workflows, longitudinal saving).

AI Features: REQ-5.7.x (Mood detection disclaimers, diarization boundaries, model prompt redaction).

Communication: REQ-5.8.x (Encrypted messaging, SOS emergency workflows, telehealth integration).

Family Access: REQ-5.11.x (Shared calendars, strict read-only family member scoping).

5.2 Test Types

Test Type	Purpose	Tools	Owner	SRS / ERD Coverage	Frequency	Tooling Cost
Unit Testing	Validate isolated functions and logic.	JUnit 5, Mockito, Flutter test	Dev Leads	All core application logic across Fig 2 ERD.	Every PR / Commit	Free / OSS
Integration / API	Verify API contracts and database IO.	REST Assured, Postman/Newman	Backend Lead	EVV REQ-1401-1404, AI Invoicing REQ-1501-1507	Nightly / On-Merge	Free / OSS
Contract Testing	Verify consumer/provider API schema agreement.	Pact, OpenAPI diff	Backend Lead	Cross-boundary API dependencies (Frontend to Backend).	On-Merge	Free / OSS
UI / E2E Testing	End-to-end user workflow validation.	Flutter integration_test, Appium	Frontend Lead	Full patient and caregiver workflows.	Pre-Release	Free / OSS
Security Testing	Identify vulnerabilities, validate encryption.	OWASP ZAP, Snyk, Trivy	QA/Security Lead	Encryption REQ-6.1, RBAC REQ-6.3, HIPAA rules.	Nightly (DAST), Every PR (SAST)	Snyk Free Tier / OSS

Test Type	Purpose	Tools	Owner	SRS / ERD Coverage	Frequency	Tooling Cost
Penetration Testing	Manual exploitation of business logic flaws.	Burp Suite Community, Manual	QA/Security Lead	Auth bypass, JWT tampering, Privilege escalation.	Monthly (pre-M3/M4)	Free Edition
Performance & Load	Verify latency and throughput thresholds.	k6, JMeter	QA/Security Lead	p95 API < 300ms, EVV < 500ms, AI < 8s.	Sprint Boundaries	Free / OSS
Compliance & Audit	Verify HIPAA controls and logging.	Custom DB scripts, log parsers	QA/Security Lead	AuditEvent and ModelPromptLog append-only validation.	Nightly	N/A
Chaos Testing	Test resilience against network failures.	Toxiproxy, Gremlin (Free)	QA/Security Lead	EVV 24-hour offline window REQ-6.4 resilience.	Pre-Release	Free Tier
Accessibility (a11y)	Verify UI meets WCAG 2.1 AA standards.	axe-core, WAVE	Frontend Lead	High contrast, screen reader compatibility.	Weekly	Free / OS

5.2.1 Test Type Deep-Dives

Security Testing: Security validation is integrated continuously. During sprint planning, we perform lightweight STRIDE threat modeling on new features. Snyk SCA scans run automatically via GitHub actions to flag vulnerable dependencies. Nightly, OWASP ZAP executes a Dynamic Application Security Testing (DAST) scan against the staging environment. Once per month, the QA/Security Lead performs a manual penetration testing session using Burp Suite Community to probe for complex business-logic flaws, such as parameter tampering in JWT tokens or horizontal privilege escalation between caregiver accounts.

Performance Testing: The CareConnect architecture relies heavily on serverless execution (AWS Lambda). Performance tests simulate realistic load spikes. Using k6 and JMeter, we run baseline, spike, and soak tests targeting the strict SLAs defined in the SRS: network round-trip < 300ms, EVV synchronous submit < 500ms, and AWS Bedrock invoice extraction < 8 seconds. We monitor Spring Boot Actuator metrics and AWS CloudWatch during execution to identify connection pool exhaustion (e.g., HikariCP limits).

Compliance & Audit Testing: This is a highly specialized test suite unique to our healthcare domain. Automated SQL assertions are run against the PostgreSQL staging database, actively attempting to execute **UPDATE audit_event SET...**

Or **DELETE FROM model_prompt_log**. The test expects these queries to throw specific database trigger exceptions, mathematically proving that the append-only integrity of the HIPAA audit trail is intact. We also run automated checks verifying that AWS KMS keys are actively attached to the RDS instance.

Offline & Chaos Testing: Requirement REQ-6.4 mandates that caregivers operating in low-connectivity areas can queue EVV actions for up to 24 hours. Using Toxiproxy, we simulate intermittent network drops, massive packet loss, and full disconnects while executing Flutter E2E tests. The tests verify that **offline_sync_queue** records are correctly written, encrypted locally, and successfully replayed to the **visit_event** table once connectivity is restored, prioritizing the earliest signature timestamp in conflict scenarios.

5.3 Manual and Automated Testing Balance

Target automation coverage: 80% Unit, 70% API, 40% E2E by Milestone 3 (07/25/2026).
table once connectivity is restored, prioritizing the earliest signature

Manual focus areas: While automation handles regression and API logic, exploratory manual testing is reserved for high-complexity human workflows. This includes evaluating AI invoice extraction confidence boundaries (REQ-1503), determining telehealth audio/video call quality under poor network conditions (REQ-5.8.6), activating SOS emergency modes under stress (REQ-5.8.3-002), and conducting tactile accessibility validation with screen readers.

Module	Manual Testing Allocation	Automated Testing Allocation
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EVV (Blue)	20% (Geofence edge cases, offline handling UX)	80% (State transitions, API endpoints, sync logic)
AI Invoicing (Purple)	40% (Extraction confidence review, visual document overlap)	60% (Pipeline triggers, OCR mock responses, workflow gates)
Communication	50% (Audio/Video quality, SOS stress tests, UI responsiveness)	50% (Message delivery APIs, WebSocket connection scaling)
Wearables	30% (Device pairing UX, edge threshold UI alerts)	70% (Data ingestion mocks, OAuth flows, database writes)
Digitization	30% (Form usability, OCR edge cases, mobile capture)	70% (Validation rules, API submissions, RBAC checks)
Identity/RBAC/Audit	10% (Exploratory bypass attempts, session timeout UX)	90% (Token lifecycle, append-only logs, controller permissions)

5.3.1 Automation Tool Stack

Test Layer	Tooling	Language	CI/CD Integration
Unit (Backend)	JUnit 5, Mockito, AssertJ	Java	Runs on every PR (Blocks merge)
Unit (Frontend)	Flutter test, Mocktail	Dart	Runs on every PR (Blocks merge)
API / Integration	REST Assured, Postman/Newman	Java / JS	Runs nightly against dev/staging
End-to-End (E2E)	Flutter integration_test,	Dart / JS	Runs pre-release (Milestone gates)
Security (SAST/DAST)	Appium 2.x OWASP ZAP, Snyk CLI, Trivy, gitleaks	N/A	SAST on PR; DAST nightly
Performance	k6, Apache JMeter	JS / Java	Manual trigger / Pre-release
Accessibility	axe-core, Pa11y	JS	Weekly scheduled runs

Test Layer	Tooling	Language	CI/CD Integration
Resilience / Chaos	Toxiproxy	CLI / API	Manual trigger / Pre-release

5.3.2 Test Data Management

High-quality testing requires robust, realistic data. We rely entirely on synthetic data generators to populate our environments. Using tools like *Faker* for demographic data, custom Python scripts to generate erratic GPS traces for EVV geofence testing, and PDF generation libraries to create "fake invoices" with complex line items, we ensure comprehensive test coverage without utilizing real PHI.

All data generation is strictly governed by the HIPAA Safe Harbor de-identification standard (§164.514). Test data is refreshed at the start of every sprint to prevent "state pollution" and test fragility. Data is strictly isolated; the Dev database never communicates with Staging, ensuring clean boundaries.

5.3.3 Environment Strategy

- **Dev Environment:** Used by engineers for local and containerized testing. Highly volatile. Seeded with minimal synthetic data sets.
- **Staging Environment:** Matches the production architecture (AWS RDS, Lambda, API Gateway). Used for automated API testing, nightly DAST scans, and QA manual exploratory testing. Contains comprehensive synthetic datasets representing thousands of patients and caregivers.
- **UAT (User Acceptance Testing) Environment:** Stable environment used exclusively for final milestone sign-off by the Instructor and Client. Upgraded only at milestone boundaries. Contains pristine, curated test records.

5.4 AI-Assisted Testing and Review

Team D leverages generative AI to accelerate test design and improve coverage, strictly within the boundaries of privacy and security constraints.

5.4.1 Approved AI Tools and Use Cases

Tool	Approved Uses	Prohibited Uses	Logging Mechanism
Claude 3.5 (Anthropic)	Test case generation from SRS REQ IDs; logic review of test scripts.	Never paste production logs, source code with secrets, or PHI.	<code>model_prompt_log</code>

GitHub Copilot	Generating boilerplate assertions (e.g., JUnit, Mockito setups) in IDE.	Never accept AI-generated test logic without human review.	IDE telemetry / PR Review
ChatGPT 4o (OpenAI)	Generating synthetic test data structures (JSON payloads, CSVs).	Never paste real patient schemas or proprietary architecture docs.	Manual entry to Appx D

5.4.2 AI Output Quality Gates

Before any AI-generated test asset (script, payload, or test case) is committed to the repository, it must pass the following human-verified checklist:

- References real, documented API endpoints.
- Traces back to a legitimate SRS REQ ID.
- Accurately references Figure 2 ERD entities (no hallucinations of tables).
- Absolutely NO PHI or PII leaked in the output or prompt.
- Assertion logic has been reviewed and logically verified by an engineer.
- Usage is documented and traceable via the model prompt log, UUID, or Appendix D.

6. Test Environment

6.1 Environment Overview

Team D will test its assigned Milestone 2 capabilities within the shared CareConnect repository and a controlled local development environment that supports backend services, integration boundaries, and related frontend workflow validation. The test environment is designed to cover Electronic Visit Verification (EVV) enhancements, AI-assisted invoicing, wearable integration boundaries, QR and invite onboarding, home care document digitization, and structured assessment workflows. Team D's planned backend stack aligns with Java 17, Spring Boot 3.4.5, Spring Data JPA, and PostgreSQL, while cloud and AI integration assumptions include AWS Bedrock, Claude, and related managed AWS services where approved. Repository testing norms also require mock-data-driven tests, shared fixtures, and avoidance of live network or database dependencies in unit testing.

Environment Area	Planned Configuration	Status	Notes
Local development and test setup	Developer workstation with access to the shared CareConnect codebase, backend modules, frontend modules, and project test suites	Ready	Local execution will use mock fixtures, sanitized records, and seeded test data instead of live production data
Repository and shared codebase	Shared CareConnect repository with backend, frontend, quality, docs, and scripts areas	Ready	Team D testing will focus on relevant APIs, service behavior, integration boundaries, and dependent UI flows
Backend runtime	Java 17, Spring Boot 3.4.5, Spring Data JPA, and PostgreSQL-backed persistence assumptions	Ready	Primary validation includes controller behavior, service behavior, field validation, persistence logic, and interface contracts
Frontend validation environment	Browser-based and workflow-oriented validation for invoice review, onboarding, document, and assessment flows that depend on backend responses	Ready	Frontend checks are included only when necessary to confirm end-to-end behavior tied to Team D features
Test fixtures and helper framework	Shared mock-data conventions and reusable fixtures under backend and frontend test-support folders	Ready	Unit tests will use reusable fixtures and helpers rather than ad hoc object creation
External integrations	AWS Bedrock/Claude, wearable APIs, EVV aggregator connections, authentication and role endpoints, document APIs, and assessment APIs	Partial / Assumption-Based	External paths will be tested through mocks, stubs, preview modes, or sandbox responses unless approved credentials and final contracts are available

Environment Area	Planned Configuration	Status	Notes
Database and migration dependencies	Shared relational persistence model and approved schema paths	Partial / Assumption-Based	Migration-dependent testing will be coordinated with approved owners to avoid validating against unapproved schema changes
AI and OCR processing path	Invoice extraction flow using mock responses or approved provider-backed behavior	Partial / Assumption-Based	Testing will validate extraction review behavior, low-confidence handling, correction workflow, and controlled failure handling even when production AI access is unavailable

6.2 Hardware, Software, and Tools

The following hardware, software, and testing tools support Team D's backend, AI, and integration responsibilities.

Category	Tool / Platform	Purpose in Testing	Owner / Primary User
Source control and shared codebase	GitHub CareConnect repository	Access shared modules, inspect affected files, coordinate implementation and testing evidence, and trace dependency changes	Team D
Backend platform	Java 17 + Spring Boot 3.4.5	Run backend services and execute backend-focused tests	Backend / AI / Integration Lead
Persistence layer	PostgreSQL + Spring Data JPA	Validate persistence behavior, schema assumptions, and record lifecycle handling	Backend / AI / Integration Lead
Cloud and AI platform	AWS Bedrock + Claude	Validate invoice AI integration behavior, response handling, and fallback logic when approved for testing	Backend / AI / Integration Lead with Team A dependency
Supporting AWS services	AWS Lambda, S3, CloudWatch, KMS, and related managed services	Support storage, observability, secure handling assumptions, and platform integration review	Shared platform dependency
Integration interfaces	EVV endpoints, wearable connector OAuth adapters, wearable metric ingestion service, invite token API, document APIs, and assessment APIs	Validate request/response behavior and interface compatibility	Backend / AI / Integration Lead
Backend test locations	/backend/core/src/test/java	Primary location for unit and backend service/controller tests	Team D

Category	Tool / Platform	Purpose in Testing	Owner / Primary User
Shared backend fixtures	/backend/core/src/test/java/com/careconnect/testsupport/fixtures	Reusable mock object generation and consistent test setup	Team D
Frontend test locations	/frontend/test and /frontend/test/test_support	Confirm backend-dependent UI behavior where end-to-end validation is needed	Frontend Lead with Team D coordination
Issue and defect tracking	Shared issue tracker and decision documentation	Track failed tests, blockers, assumptions, and resolution notes	Team Lead and Backend / AI / Integration Lead
AI assistance for documentation support	Approved AI drafting and review tools with mandatory human review	Draft candidate test cases, summarize blockers, and organize dependency notes only	Team D, human-reviewed

6.3 Test Data Requirements

All Team D testing will use synthetic, mocked, seeded, de-identified, or instructor-approved non-sensitive data only. No real patient data, live credentials, real financial content, or protected health information will be used in test records, uploaded artifacts, screenshots, or AI-supported drafting workflows. Test data must support positive-path, negative-path, authorization, low-confidence AI, malformed payload, retry, and repeated-use scenarios so that Team D can demonstrate realistic testing coverage without violating privacy or security constraints.

Test Data Set	Purpose	Source	Privacy / Handling Notes
Synthetic user accounts and roles	Validate role-based access, restricted endpoint behavior, and negative authorization cases	Seeded or mocked	No real identities; role combinations must include both permitted and non-permitted users
EVV visit records	Validate lifecycle transitions, status updates, validation rules, and preview-mode workflows	Mocked or seeded	No real provider or patient identifiers
Sample invoices and OCR extraction fixtures	Test upload, extraction review, low-confidence handling, manual correction, and approval workflows	Mocked or sanitized	No real financial or PHI data; use masked values and fictional vendors
Invite tokens and QR onboarding records	Test valid, expired, revoked, reused, and malformed token scenarios	Mocked or generated	Tokens must be test-only and time-bounded
Wearable payload samples	Test ingestion, normalization, missing-field handling, and unavailable-service behavior	Mocked or sandbox-format payloads	No live personal device data

Document and assessment form records	Validate draft, submit, review, and validation workflows	Mocked or seeded	Use fictional patient and caregiver scenarios only
Negative and fault-condition fixtures	Validate retries, provider outage handling, unauthorized access, malformed payloads, and validation failures	Mocked	Required for reliability and defensive testing
Audit and logging verification samples	Confirm redaction and safe logging practices	Test-generated	Logs must not expose PHI, tokens, secrets, or sensitive payload details

7. Roles and Responsibilities

7.1 RACI Matrix

Legend: R = Responsible (Does the work), A = Accountable (Approves/Owns the work), C = Consulted (Provides input), I = Informed (Kept up to date).

Task / Activity	Team Lead (Davon)	Backend Lead (Nhial)	Frontend Lead (Angel)	Business Analyst (Cristobal)	QA/Security Lead (Bernabe)
Test Planning	A	C	C	R	R
Test Case Design	I	R	R	C	A
Test Execution	I	C	C	I	R/A
Defect Triage	A	R	R	C	R
Security Scanning	I	C	I	I	R/A
Performance Testing	I	C	I	I	R/A
Compliance Validation	A	C	I	C	R
Release Sign-off	A	C	C	C	R
AI Test Asset Review	I	R	R	C	A
Test Environment Setup	A	R	I	I	C
Test Data Provisioning	I	R	C	C	R
Audit Log Verification	I	R	I	I	A

RBAC Enforcement Testing	I	R	R	C	A
Vulnerability Remediation	A	R	R	I	C

7.2 Detailed Responsibilities

Role	Primary Test Responsibilities	Secondary Responsibilities	Tools Owned	Reporting Cadence
QA/Security Lead (Bernabe Garcia)	E2E Test Execution, Security Scans, HIPAA Control Validation, Defect Triage.	Performance Testing, RTM Tracking, Chaos Testing.	ZAP, Snyk, k6, Postman	Daily QA Status, Weekly Bug Report
Team Lead (Davon Davis)	Milestone Sign-Off, Resource Allocation, Escalation Management.	Environment strategy alignment, Client communication.	GitHub Projects	Weekly Team Status
Backend Lead (Nhial Mayar)	Unit Testing (Java), API Contract Testing, DB schema validations.	Synthetic data generation scripts, AWS mock integrations.	JUnit, REST Assured	Per-PR Test Status
Frontend Lead (Angel Guanlao)	UI Unit Testing (Dart), E2E Flutter Tests, Accessibility compliance.	UI error state handling, Mobile device compatibility.	Flutter test, axe-core	Per-PR Test Status
Business Analyst (Cristobal Garcia)	Acceptance Criteria definition, User Story review.	Exploratory usability testing, Edge-case scenario discovery.	Confluence/Documents	Sprint Planning / Retros

8. Test Schedule and Milestone Alignment

Milestone / Period	Testing Focus	Planned Output
Milestone 2	Define test strategy, test environment, test cases, acceptance criteria, and cross-team dependencies.	Software Test Plan Technical Design Document
Milestone 3	Execute and refine feature, integration, regression, and defect retest cases as implementation matures.	Updated Software Test Plan and test evidence
Milestone 4	Complete final test execution, summarize results, document defects, and support release readiness.	Test Report

8.1 Team Testing Timeline

Date / Week	Activity	Owner	Output / Evidence
Milestone 3	Draft test plan sections	Team D	Software Test Plan
Milestone 3 start	Review SRS/TDD coverage	Team Lead	Requirements Coverage Matrix
Milestone 3 start	Create test cases	QA Lead	Test case inventory
Milestone 3 Start	Begin feature testing	Team D	Test execution evidence
Milestone 3 end	Complete regression testing	Team D	Updated test results
Milestone 4	Final validation and readiness review	Team Lead and QA Lead	Test Report

9. Test Case Design Standards

9.1 Test Case Naming and ID Convention

Use a consistent naming convention so test cases can be referenced across the TDD, Software Test Plan, defect log, and final Test Report. Suggested format: TC-[Team]-[Area]-###like TC-A-LOGIN-001 or TC-C-UI-001.

9.2 Test Case Template

Field	Description
Test Case ID	Unique identifier.
Requirement / Design Reference	SRS ID, TDD section, or feature name.
Test Case Title	Short description of the behavior being tested.
Preconditions	Required setup before executing the test.
Test Steps	Step-by-step actions.
Test Data	Inputs, accounts, mock data, device/browser, or configuration.
Expected Result	Expected system response.
Actual Result	Observed result during execution.
Status	Not Run / Pass / Fail / Blocked / Deferred.
Defect ID	Linked defect, if applicable.

9.3 Sample Test Case Row

Test Case ID	Title	Preconditions	Steps	Expected Result	Status
TC-[TEAM]-001	[Verify assigned workflow]	[System ready, user logged in, test data available]	1. [Step] 2. [Step] 3. [Step]	[Expected outcome]	Not Run

10. Requirements and Design Coverage Matrix

This matrix connects all 47 Team D requirements and non-functional requirements to their planned test cases. Requirements marked Partial have limited Milestone 2 coverage due to design-phase constraints. Full coverage is planned for Milestone 3. Other teams own requirements marked Not Covered or deferred beyond Team D's current scope.

Requirement / Design ID	Summary	Priority	Test Case ID(s)	Coverage Status
REQ-033	Invite token generation - HMAC-SHA256 signed, cryptographically random raw token	High	TC-D-INVITE-001, TC-D-INVITE-002, TC-D-INVITE-003	Covered
REQ-034	Raw token never stored - only SHA256 hash persisted in InviteToken table	High	TC-D-INVITE-002, TC-D-SEC-008	Covered
REQ-035	Invite delivered as QR code (qr_flutter) and copyable URL	High	TC-D-INVITE-003	Covered
REQ-036	Valid token validation returns invite metadata; acceptance creates user with correct role	High	TC-D-INVITE-004, TC-D-INVITE-008	Covered
REQ-037	Token error handling: expired → 410, already used → 409, revoked → 410	High	TC-D-INVITE-005, TC-D-INVITE-006, TC-D-INVITE-007	Covered
REQ-038	Admin can revoke active tokens via DELETE <code>/api/v1/invites/{id}</code>	Medium	TC-D-INVITE-007	Covered
REQ-091	Google Health Connect permission grant creates WearableConnection; denial shows guidance	High	TC-D-WEAR-001, TC-D-WEAR-002	Covered
REQ-092	Batch metric sync persists records (max 500); dashboard displays latest per type	High	TC-D-WEAR-003, TC-D-WEAR-004, TC-D-WEAR-008	Covered
REQ-093	Threshold alerts dispatched via FCM; 60-minute cooldown enforced	High	TC-D-WEAR-005, TC-D-WEAR-006	Covered
REQ-094	Consent revocation stops future syncs with CONSENT_ALREADY_REVOKED	High	TC-D-WEAR-007	Covered
REQ-095	Threshold configuration API; alert recipient IDs managed per patient/metric	Medium	TC-D-WEAR-005	Covered
REQ-1401	EVV check-in/check-out lifecycle; geofence validation; aggregator dependency dashboard	High	TC-D-EVV-001, TC-D-EVV-002, TC-D-EVV-003, TC-D-EVV-004, TC-D-AGG-001, TC-D-AGG-004	Covered

REQ-1402	Visit correction/retry workflow - EvvVisitCorrection entity with before/after snapshot	High	TC-D-EVV-005, TC-D-EVV-006	Covered
REQ-1403	Per-transition audit trail - EvvVisitAuditEntry for every state change	High	TC-D-EVV-009, TC-D-EVV-010	Covered
REQ-1404	Pre-export field validation before aggregator submission	High	TC-D-EVV-007, TC-D-EVV-008	Covered
REQ-1501	Invoice file upload (PDF/JPEG/PNG, max 10 MB with S3 SSE-KMS storage)	High	TC-D-INV-001, TC-D-INV-002, TC-D-INV-003	Covered
REQ-1502	Async Bedrock extraction invocation; status polling via GET /invoices/{id}	High	TC-D-INV-004, TC-D-INV-005	Covered
REQ-1503	Extracted fields and per-field confidence scores stored as JSONB	High	TC-D-INV-004	Covered
REQ-1504	Field correction creates InvoiceAuditEntry; reviewer continuity on re-review	High	TC-D-INV-006, TC-D-INV-007	Covered
REQ-1505	Bedrock failure routes to MANUAL_REVIEW with BEDROCK_UNAVAILABLE error	High	TC-D-INV-005	Covered
REQ-1506	Duplicate invoice detection; reviewer must acknowledge before approval	Medium	TC-D-INV-009	Covered
REQ-1507	Invoice requires explicit human approval before APPROVED status	High	TC-D-INV-008, TC-D-INV-010	Covered
REQ-1601	Document upload - W-4, I-9, Direct Deposit accepted; file stored S3 SSE-KMS	High	TC-D-DOC-001, TC-D-DOC-002, TC-D-DOC-003	Covered
REQ-1602				
REQ-1603				
REQ-1604				
REQ-1605				
REQ-1606				
REQ-1607				
REQ-1701				
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REQ-1804				
REQ-1805				
REQ-1806				

REQ-1807				
REQ-1808				
REQ-1809				
NFR-01				
NFR-02				
NFR-03				
NFR-04				
NFR-05				
NFR-06				

11. Test Execution Process

11.1 Entry Criteria

Before formal QA execution can begin on a new build, the following gates must be met:

- **Successful Deployment:** The build is successfully deployed to the Staging test environment with a passing CI pipeline (linting, unit tests, and dependency scans all green).
- **Requirement Tracing:** All requirements targeted for the sprint have at least one designed test case logged in the system and linked to the corresponding SRS REQ ID.
- **Data Seeding:** Test data is fully seeded (using synthetic data only—absolutely no PHI), covering all Figure 2 ERD entities relevant to the sprint's scope.
- **Smoke Pass:** The Build Acceptance Test (BAT) suite passes cleanly on the new build, proving core viability.
- **Environment Health:** Environment health is verified—database migrations applied, AWS KMS keys rotated if scheduled, and the audit event table is verified as reachable.

11.1.1 Build Acceptance Test (BAT)

The BAT is a critical 10-minute automated smoke pack that runs immediately upon staging deployment. If any step fails, the build is rejected and sent back to development:

1. System health endpoints (/actuator/health) return HTTP 200 OK.
2. User authentication succeeds with valid credentials and MFA prompt triggers.
3. RBAC enforcement acts correctly (a Caregiver token attempting an Admin route receives HTTP 403).
4. EVV Check-in and Check-out happy path completes successfully.
5. Invoice upload succeeds and the Bedrock callback mock returns a 200 response.
6. A secure message is successfully sent and received via WebSocket.
7. A synthetic wearable data point is successfully POSTed to the telemetry endpoint.
8. A corresponding audit_event row is written to the database for the login action.
9. An attempted UPDATE SQL command on the audit event table is violently rejected by the database trigger.
10. API Gateway handshake verifies TLS 1.3 cipher suites only (rejecting TLS 1.1/1.2).

11.2 Execution Steps

Once entry criteria are met, testing proceeds via a strict methodological sequence:

1. Pull Test Plan: QA pulls the latest test plan and assigned test cases from the GitHub Issues registry, ensuring alignment with the sprint scope.
2. Seed Environment: Reset and re-seed the PostgreSQL database with sprint-specific synthetic data to ensure a clean state and prevent test-data cross-pollution.
3. Automated Regression: Execute the full automated regression suite. The performance target for this suite is completion in under 35 minutes to preserve fast feedback loops.
4. Exploratory Charters: Execute manual exploratory charters targeting complex human elements (e.g., simulating a user rapidly crossing an EVV geofence boundary, or testing the limits of AI invoice extraction confidence).
5. Log Results: Document pass/fail status meticulously. Failed tests must include attached screenshots, HAR files, or server logs.
6. File Defects: File discovered defects in GitHub Issues using the standardized defect template (Section 12.2). Label them accurately by severity and affected modules (EVV, Invoicing, Comm, Wearable, Identity, Audit).
7. Update RTM: Update the Requirements Traceability Matrix (RTM) daily to reflect which REQ IDs have passed their associated tests.
8. Nightly Security Scans: Run security scans nightly (Snyk SCA on the repo, ZAP DAST against the live staging environment, Trivy on Docker containers).
9. Daily Reporting: Generate the daily QA status report (pass rates, blocker identification, security findings) and broadcast to the team via Slack.
10. Defect Triage: Hold bi-weekly defect triage meetings (Tues/Thu) to classify, prioritize, and assign outstanding bugs with the Dev Leads and BA.

11.2.1 Daily QA Cadence

The standard day begins at 9:00 AM with the team standup. Immediately following, the automated suite kickoff runs against the overnight build. By 11:00 AM, the QA lead triages any failed automated tests and assigns defects. The afternoon is dedicated to executing manual test charters, focusing on newly merged features. By 4:00 PM, the daily QA status is compiled and posted to Slack. At End of Day, the automated security and DAST scans are scheduled to run overnight.

11.2.2 Sprint Test Cycle

Testing activities follow the weekly Agile-Scrum rhythm. Day 1: Sprint planning and threat modeling. Days 2-4: Test case design and automation script authoring alongside development. Day 5: Main test execution and team "bug bash" session. Day 6: Full regression testing to ensure no collateral damage from bug fixes. Day 7: Sprint retrospective and final RTM update closure.

11.2.3 Test Case Bank — Sample Catalog

TC ID	Title	Module	Linked REQ	ERD Entity	Type	Priority	Automation
TC-E VV-001	Valid GPS check-in saves correctly	EVV	REQ-1402	visit_event	Functional	High	Automated
TC-E VV-002	Offline queue stores payload encrypted	EVV	REQ-6.4	offline_sync_queue	Resilience	High	Automated
TC-E VV-003	Out-of-geofence visit triggers supervisor alert (BR-002)	EVV	REQ-1402	geo_coordinate	Functional	Medium	Automated
TC-I NV-001	Bedrock/Claude extraction returns required fields with confidence scores	AI Invoicing	REQ-1503, REQ-1506	extraction_result	Functional	High	Automated
TC ID TC-I NV-002	Title Invoice cannot be accepted without explicit human approval	Module AI Invoicing	Linked REQ REQ-1507	ERD Entity invoice , reviewer_assignment	Type Compliance	Priority Critical	Automation Automated

TC ID	Title	Module	Linked REQ	ERD Entity	Type	Priority	Automation
TC-I NV-003	AWS Bedrock credentials are pulled from Secrets Manager (no hardcoded keys)	AI Invoicing	REQ-1502	extraction_request	Security	High	Automated
TC-AU DIT-001	Direct SQL UPDATE on audit_event is rejected by DB trigger	Audit	HIPAA §164.312(b)	audit_event	Security / Compliance	Critical	Automated
TC-AU DIT-002	Hash-chain verification detects tampered audit row	Audit	HIPAA §164.312(c)(1)	audit_event (before_hash/after_hash)	Security / Compliance	Critical	Automated
TC-CO MM-001	SOS activation immediately notifies authorized caregivers with location	Communication	REQ-5.8.3-002, REQ-5.8.3-003	conversation , message	Functional	Critical	Automated + Manual
TC-W EAR-001	Stale wearable (no sync 24h) emits user notification	Wearables	REQ-095	wearable_data_point	Functional	Medium	Automated

TC ID	Title	Module	Linked REQ	ERD Entity	Type	Priority	Automation
TC-D OC-0 01	Family Read-Only role cannot POST to document submission endpoint	Documents	REQ-1607	user , role , permission	Security / RBAC	High	Automated
TC-S CREE N-00 1	Patient role is denied access to screening submission API	Screening	REQ-1706	user , role	Security / RBAC	High	Automated
TC-A SSES S-00 1	Comprehensive assessment retains historical versions for longitudinal review	Assessment	REQ-1808, REQ-1809	care_plan (versioned)	Functional	Medium	Automated
TC-AI-00 1	Ask AI response renders mandatory medical-judgment disclaimer	AI Features	REQ-5.7.2-002	model_prompt_log	Compliance	High	Automated
TC-F AM- 001	Family Read-Only user cannot edit, submit, or approve records	Family / Social	REQ-5.11.2-002	user , role , permission	Security / RBAC	High	Automated

11.3 Exit Criteria

Test execution for a sprint or milestone is considered complete only when every gate below is met:

- 100% of planned test cases executed; **≥ 95% pass rate**.
- **Zero open Sev-1 or Sev-2 defects** on any path touching PHI handling, RBAC, EVV submission, AI invoice human-approval gate (REQ-1507), or audit logging.
- Code coverage targets met: **≥ 80% unit on backend, ≥ 70% on frontend**.
- All HIPAA control tests pass: AES-256 at rest, TLS 1.3 in transit, append-only, PHI redaction in logs and AI prompts (REQ-5.7.2-005), MFA enforced on administrator accounts.
- Performance gates met at p95: API < 300ms, EVV submit < 500ms, AI extraction round-trip < 8s, telehealth call setup < 2s.
- Requirements Traceability Matrix shows every in-sprint REQ ID covered by ≥ 1 test case.
- Security scan results: zero Critical Snyk findings, zero High ZAP findings, zero High Trivy findings on production container images.
- Sign-off recorded by QA/Security Lead, Team Lead, and Instructor for each milestone deliverable.

EVV Workflow Note:

Exit criteria explicitly require successful simulation of the EVV offline queue ([offline_sync_queue](#)) reaching the 24-hour boundary and successful state transition of [visit](#) from [scheduled](#) → [in_progress](#) → [completed](#) → [exported](#) via the [edi_export_batch](#) aggregator path (REQ-1404).

11.3.1 Quality Gates by Milestone

Milestone	Due Date	Coverage Target	Critical Defects	Performance	Security Scan	HIPAA Controls	RTM Coverage
M1 — Project Plan + SRS	05/30/2026	N/A (planning)	N/A	N/A	Tool inventory complete	Identified & mapped	Initial RTM draft
M2 — Tech Design	06/13/2026	≥ 40% unit	0 Sev-1	Baseline established	Snyk + Trivy	≥ 50% control	≥ 70% of

+ Test Plan v1					onboarded	tests authored	in-scope REQs
M3 — Program Manager Guide + Test Plan v2	07/25/2026	≥ 80% unit / 70% FE	0 Sev-1, 0 Sev-2	p95 gates met	Clean nightly ZAP scan	100% control tests pass	≥ 95%
M4 — User Guide + Test Report	08/04/2026	Maintained from M3	0 Sev-1, 0 Sev-2	Soak + spike tests passed	Pen-test complete, no Highs	100% pass; audit hash-chain verified	100%

11.3.2 Go/No-Go Decision Process

Each milestone closes with a formal Go/No-Go sign-off meeting chaired by the Team Lead and facilitated by the QA/Security Lead. Attendees include all five contributors and (for M4) Dr. Assadullah. The QA Lead presents: the RTM coverage report, the defect dashboard (open by severity), the latest security scan summary, the HIPAA control test results, and the performance baseline. A unanimous "Go" requires all 11.3 exit criteria to be met; any "No-Go" item triggers a remediation plan with named owners and a target re-review date within 48 hours.

12. Defect Management Process

12.1 Defect Severity Definitions

Severity	Definition	Examples	SLA Response	SLA Fix
Sev-1 Critical	System failure, data loss, or severe security/compliance breach.	PHI exposure in logs/URLs, auth bypass, EVV non-submission, audit_event write failure, REQ-1507 human-approval gate bypass, TLS downgrade.	1 hour	24 hours
Sev-2 High	Major functional impairment with no workaround; compliance risk.	RBAC bypass on non-PHI (REQ-1706/1809), wearable threshold alert miss (REQ-093), offline sync data corruption (REQ-6.4), missing AI disclaimer (REQ-5.7.2-002).	4 hours	3 business days
Sev-3 Medium	Moderate functional issue; workaround exists; minor degradation.	Incorrect UI state, non-PHI logging gap, performance regression < 25%, missing geofence flag visualization, confidence indicator missing on one field (REQ-1506).	1 business day	Within sprint
Sev-4 Low	Minor impact, cosmetic issues, or enhancements.	Cosmetic glitches, typos, minor WCAG warnings, slow load < 10% degradation.	3 business days	Next sprint

12.1.1 Priority vs Severity Matrix

Severity describes *technical impact*; Priority describes *business urgency*. The matrix below determines work order in the sprint backlog.

	P1 — Immediate	P2 — High	P3 — Normal	P4 — Low
Sev-1	Drop-everything	Same-day fix	Same-sprint	Same-sprint
Sev-2	Same-day fix	3-day fix	Same-sprint	Next sprint
Sev-3	Same-sprint	Same-sprint	Next sprint	Backlog
Sev-4	Same-sprint	Next sprint	Backlog	Backlog

12.1.2 Compliance-Impacting Defect Auto-Escalation

Automatic Severity Override:

The following defect categories are automatically classified as Sev-1 or Sev-2

regardless of their functional impact, due to direct regulatory exposure:

- Any PHI exposure in logs, URLs, error messages, page titles, or AI prompts (HIPAA §164.502).
- Any audit event or model prompt log integrity issues, including failed writes, mutations, or hash mismatches (HIPAA §164.312(b)).
- Any RBAC bypass — even read-only escalation (HIPAA §164.312(a)(1)).
- Any prompt-redaction filter failure prior to Bedrock/Claude submission (REQ-5.7.x).
- Any encryption-in-transit downgrade below TLS 1.3 (HIPAA §164.312(e)(1)).
- Any failure in a BAA-covered integration security boundary (AWS, SMS/email vendor, aggregator).
- Any append-only DB trigger that fails to reject UPDATE/DELETE attempts.

12.2 Defect Log Template

Every defect filed in GitHub Issues must use the standard tabular template below. Each column captures information required for traceability, prioritization, and resolution tracking across the QA/Security workstream.

Defect ID	Related Test Case	Description	Severity	Owner	Status	Resolution Notes
DEF-001	TC-[TEAM]-001	[Description]	High / Medium / Low	[Name]	Open / In Progress / Fixed / Deferred	[Notes]

Column Definitions:

- **Defect ID:** Unique sequential identifier (e.g., DEF-001, DEF-002).
- **Related Test Case:** The Test Case ID that exposed the defect, following the **TC-[MODULE] - [NUMBER]** convention (e.g., TC-EVV-001, TC-INV-002, TC-AUDIT-001).
- **Description:** A concise, one-to-two sentence summary of the observed defect behavior.

- **Severity:** Technical impact classification — **High** (system failure, data loss, security/compliance breach), **Medium** (major functional impairment with workaround), or **Low** (minor or cosmetic issue).
- **Owner:** Team member currently assigned to investigate or remediate the defect.
- **Status:** Current lifecycle state — **Open** (newly logged), **In Progress** (being actively worked), **Fixed** (resolved and verified), or **Deferred** (acknowledged but postponed beyond the current milestone with justification).
- **Resolution Notes:** Free-form notes documenting root cause, fix approach, commit r

12.3 Retest and Closure

Process flow: Fix submitted → CI passes → Deployed to Staging → QA verifies fix on the failing test case → Regression run on adjacent code paths → Status set to Verified → Closed after 1 sprint with no recurrence.

Reopen policy: Any defect that recurs within 2 sprints is automatically reopened and root cause is re-examined; a preventive-action ticket is filed.

Audit-related closures require an additional check: the QA Lead verifies that the audit event row exists for the corrective action and that no log redaction occurred.

12.4 Defect Metrics Dashboard

The QA Lead publishes a weekly metrics report. Targets are tracked against the following KPIs:

Metric	Definition	Target	Owner
Open defects by severity	Total open Sev-1/2/3/4 at week close	0 Sev-1, ≤ 2 Sev-2, ≤ 8 Sev-3	QA Lead
MTTD (Mean Time To Detect)	Hours from defect introduction (commit) to discovery	< 24h for Sev-1/2	QA Lead
MTTR (Mean Time To Resolve)	Hours from triage to verified fix	Sev-1 < 24h, Sev-2 < 72h	Module Owner
Defect density	Defects per 1k LOC, per module	< 5/kLOC	Module Owner
Escape rate	% of defects discovered after release to UAT	<5%	QA Lead

Metric	Definition	Target	Owner
Reopen rate	% of closed defects reopened within 2 sprints	<8%	QA Lead

12.5 Root Cause Analysis Categories

Category	Definition	Example	Preventive Action
Code Defect	Bug in application logic introduced by developer.	Off-by-one in geofence radius calculation.	Code review checklist, additional unit tests.
Requirements Gap	Behavior was not specified in SRS or was ambiguous.	Behavior of expired invite token not specified beyond REQ-037.	BA + QA review of acceptance criteria pre-sprint.
Test Coverage Gap	Feature behavior was specified but not tested.	No negative test for offline sync after 24h.	RTM gap audit each sprint close.
Environment Issue	Defect caused by environment drift, missing seed data, or config.	Staging KMS key not rotated, causing AES failure.	IaC parity check, weekly env diff.
Third-Party Integration	Failure originated in a vendor service (Bedrock, aggregator, wearable API).	Bedrock returns 5xx, no retry handling.	Contract tests + circuit breakers + chaos test coverage.
AI Hallucination	AI-generated artifacts (test, code, data) referenced non-existent behavior.	AI-generated test asserted on a deleted endpoint.	Apply 5.4.2 quality gates; reject and re-prompt.
Compliance Misconfiguration	Compliance control was disabled or weakened, often inadvertently.	Audit trigger dropped during a migration.	Compliance-as-code assertions in CI; weekly HIPAA check.

13. Cross-Team Integration Testing

13.1 Integration Points

Team D's Milestone 2 scope crosses several team and platform boundaries, especially in authentication, AI-assisted invoicing, EVV readiness, onboarding, document workflows, assessments, and wearables. Cross-team integration testing will therefore focus on interface compatibility, request and response contract validation, role enforcement, mock fidelity, and dependency tracking for unresolved integrations. This approach allows Team D to verify its responsibilities without introducing rework or risk from unapproved live connections.

Integration Point	Teams / Owners Involved	What Will Be Tested	What Might Break or Conflict	Coordination Method
Auth and role endpoints protecting Team D features	Team D and Team A / platform owners	Access control for invoice, assessment, document, invite, and EVV-related actions	Incorrect role mapping, hidden authorization failures, inconsistent permissions across routes	Shared decision log, coordination meetings, and issue tracking
Invoice status API and AI extraction workflow	Team D, Team A, and Team E as needed	Upload-to-review flow, extraction response handling, confidence/failure states, and approval workflow	Provider contract mismatch, response-shape changes, unsafe prompt data handling, or duplicate OCR patterns	Shared document review, dependency tracking, and decision updates
EVV write endpoints and aggregator readiness path	Team D, shared integration owners, and stakeholders	Validation, save and update flow, preview-mode submission behavior, and blocked/live boundary handling	Unknown aggregator target, missing credentials, or incompatible payload assumptions	Decision tracking and escalation log
Invite token API and onboarding workflow	Team D and Team A	Token generation and validation, expiry, revocation, reuse handling, and role assignment flow	Incorrect token-state handling, consent mismatch, or broken onboarding path	Shared coordination notes and interface review
Wearable connector OAuth adapters and metric ingestion service	Team D, platform owners, and external API boundary owners	Mock connector authorization, payload ingestion, normalization, and disconnected-service handling	Unsupported live tokens, contract drift, or missing/malformed metric fields	Mock-boundary documentation and interface review
Document Storage API and Document status API	Team D and platform services	Create, retrieve, validate, and status-track document workflows	Storage assumptions, metadata mismatch, or access-control gaps	Shared artifact and route review

Integration Point	Teams / Owners Involved	What Will Be Tested	What Might Break or Conflict	Coordination Method
Assessment API and assessment/patient profile API	Team D and data/platform owners	Draft-save-submit lifecycle, validation rules, schema expectations, and linked profile behavior	Migration conflicts, field-model mismatch, or incomplete data contracts	Shared schema review and decision updates

13.2 Shared Coordination Questions

Because several Team D responsibilities depend on unresolved or partially resolved external decisions, the following coordination questions must remain visible during testing and documentation. These questions help prevent Team D from validating against outdated assumptions or incomplete contracts.

Coordination Question	Affected Teams / Owners	Why It Matters for Testing	Current Working Assumption
Which EVV aggregator or submission target is officially approved for demonstration and later production alignment?	Team D, stakeholders, and integration owners	Determines whether EVV is tested as preview-only, sandbox-connected, or fully mocked	Test EVV external submission in preview or mock mode only until the target is approved
Who provides the approved Bedrock / Claude contract and access path for invoice AI extraction?	Team D, Team A, and Team E	Needed to confirm final request/response shape and provider-specific failure handling	Use controlled mocks or approved non-production responses when live access is unavailable
What is the final RBAC matrix for Team D-owned workflows?	Team D and Team A	Affects authorization tests, role-negative tests, and acceptance evidence	Only explicitly permitted users may access restricted functions; unresolved mappings remain documented assumptions
Which database or schema changes are approved for assessment and document workflows?	Team D and database/platform owners	Prevents testing against unapproved fields or migrations	Test shared and approved schema paths only; track blocked migration-dependent cases
Which wearable providers or sandbox payloads are officially in scope for Milestone 2?	Team D and external API owners	Prevents over-testing unsupported live integrations	Test normalized mock payloads and boundary behavior only
What is the final ownership split for OCR, document extraction, and adjacent AI responsibilities?	Team D and Team E	Prevents duplicated implementation and conflicting test evidence	Use Team D workflow boundaries and escalate overlaps through the decision log

13.3 AI Automation Support for Coordination

AI-assisted support may be used to help Team D organize and refine test work, but it does not replace human review, test execution, defect analysis, or final team judgment. Acceptable support activities include drafting candidate test cases, identifying likely coverage gaps, organizing dependency notes, summarizing coordination decisions, and comparing mocked contracts against expected interface shapes. Every AI-generated output must be reviewed, corrected where necessary, and approved by a Team D member before inclusion in official deliverables or shared coordination artifacts.

No protected health information, live credentials, secrets, or other unauthorized sensitive content may be uploaded into AI tools. If an integration remains unresolved, the official documentation must record the item as an assumption, mock-only boundary, or blocked dependency rather than present AI-generated text as confirmed implementation truth.

14. Accessibility, Usability, and Compatibility Testing

14.1 Accessibility Testing

Accessibility Area	Planned Check	Applies?	Notes
Keyboard navigation	Verify key workflows can be completed without mouse input in web contexts (EVV detail, invoice review, onboarding flow, form save/submit, assessment completion).	Yes	Planned manual walkthrough and focus-order check.
Screen reader labels	Verify form fields, action buttons, status chips, and error regions have clear accessible labels.	Yes	Prioritize invoice review fields and multi-section assessment forms.
Color contrast	Verify text, status indicators, validation messages, and action controls are readable against background.	Yes	Planned contrast checks on EVV states, alert badges, and error text.
Error messages	Verify validation and failure messages are specific, actionable, and non-technical.	Yes	Includes token errors (invalid/expired), extraction failure messaging, and form-field errors.
Captions / transcripts / alternative input	Verify non-voice alternatives exist where needed and no Team D workflow requires voice-only input.	Yes	Team D scope is non-voice-first; document any cross-team dependency exceptions.

14.2 Usability Testing

Usability review will focus on high-impact Team D workflows where errors or confusion create operational risk. Reviews will evaluate clarity, completion rate, error prevention, and recovery experience.

Planned workflow usability checks:

1. EVV visit lifecycle progression with correction/retry.
2. Invoice upload, AI extraction review, correction, and approval.
3. QR/invite onboarding including invalid/expired/used token states.
4. Wearable link/sync/alert acknowledgment flow.
5. Home-care document digitization with draft/submit/review states.
6. Screening and comprehensive assessment draft/resume/submit flow.

Usability evaluation focus:

1. Can users identify the next required action quickly?
2. Are validation and failure states understandable without technical knowledge?
3. Can users recover from mistakes without re-entering all data?
4. Are workflow transitions and completion states obvious?

14.3 Compatibility Testing

Compatibility testing is planned for the Team D UI workflows across standard browser and viewport targets used for milestone demonstrations and development validation.

Platform / Device / Browser	Test Purpose	Owner	Status
Chrome (desktop)	Validate baseline behavior for all Team D workflow screens.	Team D QA/Frontend	Planned
Edge (desktop)	Validate browser compatibility for forms, status views, and routing behavior.	Team D QA/Frontend	Planned
Firefox (desktop)	Validate rendering and workflow completion consistency.	Team D QA/Frontend	Planned
Safari (macOS)	Validate compatibility for web-based workflow execution and form interactions.	Team D QA/Frontend	Planned
Mobile viewport (responsive)	Validate usability of onboarding, document forms, and assessment interactions.	Team D QA/Frontend	Planned
Tablet viewport (responsive)	Validate workflow readability and interaction spacing across medium layouts.	Team D QA/Frontend	Planned

15. Security, Privacy, Performance, and Reliability Testing

15.1 Security and Privacy Testing

Security and privacy testing for Team D will focus on access control, safe handling of test data, controlled AI usage, secure integration boundaries, and correct behavior when tokens, uploads, documents, or external-service paths are invalid or unavailable. Because Team D's assigned features touch invoices, document workflows, onboarding tokens, assessments, EVV records, and wearable-related data, security testing must confirm least-privilege access, no unauthorized disclosure, and no unsafe use of AI-support mechanisms.

Security / Privacy Area	Test Objective	Expected Result
Authentication and RBAC	Confirm only permitted users can access restricted Team D functions and endpoints	Unauthorized users are denied access, and permitted users reach only the actions allowed by role
Invite token and QR onboarding protection	Confirm token validation enforces valid, expired, revoked, reused, and malformed-token behavior	Invalid or expired tokens fail safely with no unintended account linking or privilege assignment
Invoice upload and review protection	Confirm uploaded invoice artifacts and extraction results are visible only to authorized users	Unauthorized access is blocked and protected content is not exposed in the UI, logs, or error responses
AI prompt and data minimization	Confirm no protected or unauthorized data is sent to AI tools and only approved or sanitized test content is used	AI-related testing uses only approved mock or sanitized data and follows privacy constraints
Logging and audit safety	Confirm logs do not expose PHI, secrets, tokens, or full sensitive payloads	Logs contain only safe operational information or redacted values
Wearable and external integration boundary protection	Confirm mocked or sandbox integration paths do not bypass authentication, validation, or consent assumptions	External-boundary tests preserve security controls even when the provider is mocked
Document and assessment data handling	Confirm forms, drafts, and submissions are protected by role checks and safe storage assumptions	Sensitive workflow state is not exposed to unauthorized users and validation failures are safely handled
Secret and credential handling	Confirm no live production secrets are embedded in tests, fixtures, screenshots, or submission artifacts	Only test credentials, placeholders, or approved non-production values are used

15.2 Performance Testing

Performance testing for Milestone 2 will verify that Team D workflows respond consistently under normal development and classroom demonstration conditions. The objective is to confirm timely user feedback, stable backend behavior, and reasonable responsiveness for mocked or controlled integrations without requiring production-scale load testing. Any repeated delay, hang, or unstable workflow behavior will be treated as a defect, risk, or documented limitation.

Workflow / Component	Performance Check	Method	Pass Condition
EVV create, update, and review actions	Measure response behavior for create, update, and status-transition operations	Repeated API or UI execution under normal team test conditions	Requests complete without timeout, stale state, or duplicate record creation
Invite token validation and onboarding	Measure time to validate token state and return onboarding outcome	Repeated valid and invalid token tests	User receives prompt success or controlled failure response without hanging or unclear state
Invoice upload and extraction review initiation	Measure time to accept upload and present extraction status or review state	Repeated mock or provider-controlled workflow tests	Upload acknowledgment and review state appear in an acceptable user-facing timeframe
Assessment draft save and submit	Measure responsiveness of save, resume, and submit behavior	Repeated save and submit cycles with realistic form sizes	No noticeable blocking, state loss, or failed transitions
Document status retrieval	Measure list, detail, and status retrieval behavior	Repeated access to document-related workflows	Document state is returned consistently without intermittent failure
Wearable payload ingestion (mock)	Measure stability of mock ingestion and normalization behavior	Replay representative payload fixtures	Normalized results are processed consistently without service crash or malformed output

15.3 Reliability Testing

Reliability testing will confirm that Team D workflows remain stable when faced with repeated use, invalid input, provider unavailability, token problems, interrupted sessions, and mock-to-real boundary differences. Particular focus will be placed on preserving workflow state, preventing duplicate or partial processing, and ensuring that failure handling is controlled, understandable, and recoverable.

Reliability Scenario	What Will Be Checked	Expected Result
AI or OCR provider unavailable or mock fallback required	System behavior when extraction service is unavailable, slow, or returns an error	Workflow fails safely, preserves review context where applicable, and provides a controlled fallback or retry path
EVV external submission not approved	Behavior when live aggregator submission is blocked or unavailable	System remains in preview or mock-safe mode and does not attempt unsafe live submission
Expired, revoked, or reused invite token	Repeated token handling across edge cases	Each invalid token state is handled correctly and does not create inconsistent onboarding results
Repeated invoice review actions	Multiple corrections, approvals, or returns in the same workflow	No duplicate approvals, broken status progression, or lost edits

Assessment draft interrupted mid-workflow	Save and resume behavior after interruption, timeout, or navigation away	Draft state is preserved when intended, and incomplete submissions do not corrupt records
Wearable sync errors or malformed payloads	Handling of missing fields, bad payloads, or disconnected sources	Errors are controlled, logged safely, and do not corrupt stored data
Validation failures on documents and assessments	Server and client rejection of incomplete or invalid input	Errors are specific, actionable, and do not partially commit invalid data
Repeated execution of the same workflow	Stability across multiple runs of the same user journey	Workflow remains consistent over repeated use without hidden state corruption

16. Regression and Acceptance Testing

16.1 Regression Testing

Regression testing is planned to ensure existing or previously stabilized workflow behavior remains intact after Team D feature updates and cross-team integration changes.

Regression Area	Why It Matters	Test Case ID	Status
EVV lifecycle transitions	Broken transitions can block visit completion and downstream readiness.	TC-TD-REG-001	Not Run
EVV correction/retry path	Regression here can cause unrecoverable workflow errors.	TC-TD-REG-002	Not Run
Invoice review/edit/approve flow	Preserving reviewer continuity is core to migration success.	TC-TD-REG-003	Not Run
Onboarding token validation handling	Incorrect token logic creates account-linking failures and support burden.	TC-TD-REG-004	Not Run
Wearable status display and alert state handling	Regressions can hide critical sync/threshold information.	TC-TD-REG-005	Not Run
Document draft/submit/review lifecycle	Broken lifecycle states undermine operational traceability.	TC-TD-REG-006	Not Run
Screening/assessment draft-resume-submit flow	Regressions risk clinical data loss and incomplete assessments.	TC-TD-REG-007	Not Run

16.2 Acceptance Criteria

Acceptance criteria below are milestone-appropriate and evidence-driven, but currently remain in planned status until execution occurs.

Acceptance Criterion	Evidence Required	Owner	Status
Team D workflows implement mapped SRS requirement IDs for 5.1–5.6 scope.	Requirement-to-test mapping and walkthrough evidence.	Team D Lead/BA	Planned
Core UI workflows complete end-to-end without blocking defects in the baseline environment.	[Manual test results and screenshots/videos.	Team D QA	Planned
Error/alternate flows present clear, actionable user guidance.	Validation and failure-state evidence set.	Team D QA/UX	Planned
Cross-team dependencies are documented with current status and assumptions.	Updated STP/TDD dependency sections.	Team D Lead	Planned
Regression checklist covers critical pre-existing workflows affected by Team D changes.	Completed regression checklist and outcomes.	Team D QA	Planned

16.3 Readiness Decision

Current readiness status: Partially Ready (Planning Complete, Execution Pending).

Reason:

1. Test design, coverage intent, and dependency mapping are documented.
2. Test execution evidence is not yet available.
3. Several integration points remain dependency-sensitive (for example live external integrations and cross-team contracts).

Next action:

1. Execute prioritized UI and regression test cases in milestone test window.
2. Capture evidence artifacts (screenshots/logs/video notes) per acceptance criterion.
3. Update readiness to Ready / Blocked / Deferred based on observed outcomes.

17. Test Deliverables

Deliverable	Description	Milestone Use
Software Test Plan (this document)	Defines the complete planned testing approach, 80 test cases, coverage matrix, risks, cross-team dependencies, and acceptance criteria for Team D's 8 feature areas.	Milestone 2 submission
Test Case Set (Appendix A)	Full inventory of 80 test cases plus detailed step-by-step test cases for all 80 planned cases, organized by 12 test areas.	Milestone 2 and Milestone 3
Requirements Coverage Matrix (Section 10)	Maps all 47 Team D requirements and NFRs to specific test case IDs with coverage status (Covered/Partial/Not Covered)	Milestone 2 and beyond
Test Execution Log (Appendix B)	Date-stamped record of each test case execution including tester, result, evidence location, and notes. Populated starting Milestone 3.	Milestone 3 and Milestone 4
Defect Log (Appendix C)	Tracks all defects found during testing with DEF-D-### IDs, severity, owner, status, and resolution notes. Populated starting Milestone 3.	Milestone 3 and Milestone 4
Test Evidence Package	Screenshots, CI log exports, CloudWatch evidence, Postman collection results, and Flutter test output confirming test execution results.	Milestone 3 and Milestone 4
Final Test Report	Separate Milestone 4 deliverable summarizing executed test results, defect counts by severity, acceptance criteria outcome, and release readiness decision.	Milestone 4

18. Risks, Assumptions, Constraints, and Dependencies

18.1 Risks

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
R-QA-01	AWS Bedrock / Claude integration availability (Under Review in Project Plan).	Medium	High	Mock Bedrock client; record-and-replay extraction fixtures; feature flag.	QA/Security Lead
R-QA-02	EVV aggregator access / documentation gaps (At Risk in Project Plan).	High	High	Schema-only contract tests; synthetic EDI 837 fixtures; document-only delivery per REQ-1404.	Backend Lead
R-QA-03	Wearable API rate limits or sandbox revocation (Fitbit / HealthKit / Health Connect).	Medium	Medium	API stubs and contract tests; record-replay mode for REQ-091-095.	QA/Security Lead
R-QA-04	PHI accidentally enters AI prompts during	Low	Critical	Prompt-redaction middleware + <code>model_prompt_log</code> review; automated regex/PII scanner	QA/Security Lead

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
	testing or production.			pre-commit; AI-prompt review checklist (5.4.2).	
R-QA-05	Test environment drift from production-like configuration.	Medium	Medium	IaC parity, weekly environment diff check, KMS key rotation parity.	Team Lead
R-QA-06	Team availability — coursework and external responsibility conflicts.	High	Medium	Cross-training matrix (Section 7.4), automation-first strategy.	Team Lead
R-QA-07	Insufficient negative test data for geofence and offline 24h scenarios.	Medium	High	Chaos scripts and synthetic GPS traces; BA-authored scenario expansion.	QA/Security Lead
R-QA-08	Append-only audit table compromise (UPDATE/DELETE bypass).	Low	Critical	DB trigger preventing UPDATE/DELETE on <code>audit_event</code> and <code>model_prompt_log</code> ; nightly hash-chain verification.	Backend Lead
R-QA-09	Snyk free-tier scan limits exceeded mid-capstone.	Medium	Medium	Schedule scans intelligently; prioritize critical repos; fall back to OWASP Dependency-Check.	QA/Security Lead

Risk ID	Description	Likelihood	Impact	Mitigation	Owner
R-QA-1 0	AWS test environment cost overrun (unexpected charges).	Low	Medium	Auto-shutdown of non-prod resources nights/weekends; budget alerts.	Team Lead
R-QA-1 1	Insufficient diversity in synthetic test data leading to missed bias in AI extraction.	Medium	Medium	Partner with BA for scenario expansion; include multilingual + edge-format invoices.	Business Analyst
R-QA-1 2	Wearable data source revocation or API breaking change mid-capstone.	Low	High	Mock at OAuth layer; pin SDK versions; document a re-onboarding runbook.	QA/Security Lead
R-QA-1 3	Late delivery of ERD/Design from Backend	Medium	High	Schema-first contract testing using Figure 2 ERD. Freeze schema by end of Sprint 3.	Team Lead
R-QA-1 4	Lead delays integration test design. HIPAA control test failure discovered late blocks M4 sign-off.	Low	Critical	Frontload all compliance tests to M2; weekly control-test dashboard from M2 onward.	QA/Security Lead

18.2 Assumptions

Team A CI/CD ownership: Team A delivers and maintains the shared CI/CD pipeline including SAST, SCA, container scanning, and signed releases. Our test gate layer above this baseline.

AWS service availability: AWS KMS, Secrets Manager, Bedrock, S3, and RDS PostgreSQL are provisioned and accessible in Dev and Staging by Sprint 2.

Instructor approval of synthetic-data approach: Use of Faker-based PHI generators following HIPAA Safe Harbor (§164.514(b)) is approved for academic capstone use.

EVV aggregator stub: Even if live aggregator access is unavailable, a documented schema and a stub endpoint will be available by Sprint 4 to satisfy REQ-1404 testing.

Technology stack stability: Spring Boot 3.4.5 / Spring Security and Flutter+Dart stacks remain stable through the capstone window.

ERD authoritativeness: Figure 2 ERD (Sections 7.2-7.6 of the Technical Design Document) reflects the as-implemented schema as of M2 and changes thereafter are version-controlled.

Tool approval: The instructor approves use of public OWASP testing tools (ZAP, Dependency-Check), Snyk free tier, Trivy, and Claude/Copilot for AI-assisted testing.

Repository-level testing precedes E2E: All Figure 2 ERD entities will have repository-level integration tests authored by Backend Lead before QA writes E2E tests on top.

Environment access: Test environment access is granted to all 5 team members by Sprint 2, with role-appropriate IAM policies.

Pipeline parity: The Team A CI/CD pipeline reaches feature parity with our test gates by M2 so that QA gates are enforceable on every PR.

18.3 Constraints

Fixed SWEN 670 capstone calendar (M1 → M4 by 08/04/2026); no flexibility on milestone dates.

No real PHI permitted in any environment — including Dev, Staging, or UAT — under any circumstance.

No legally binding e-signatures or custom hardware development permitted in the project scope.

Limited budget for commercial scanners; rely on OSS / community editions (OWASP ZAP, Snyk free tier, Burp Community, Trivy, OWASP Dependency-Check).

Team members balance coursework and external responsibilities; QA Lead effort budget is approximately 10–15 hours/week.

Maximum sandbox storage for synthetic data: 50 GB on an AWS S3 test bucket; older fixtures rotated quarterly.

Network egress restriction: No outbound traffic from test environments to public internet except whitelisted vendor endpoints (Bedrock, KMS, wearable APIs, aggregator stub).

Total documented QA effort budget: approximately 120 hours over the capstone period (M1 through M4) split across all five contributors with QA Lead carrying ~60 hours.

18.4 Dependencies

Upstream technical: Team A CI/CD, AWS Bedrock/Claude, AWS KMS & Secrets Manager, EVV aggregator API (or stub), wearable OAuth providers.

Internal: Backend Lead delivers stable API contracts each sprint; Frontend Lead delivers a buildable test app; BA delivers acceptance criteria with REQ IDs and ERD entity references.

Documentation: Approved Technical Design Document Sections 7.2-7.6 (Data Model) and Figure 2 ERD before integration test design begins.

External: Vendor BAAs in place for AWS, SMS/email providers, and the aggregator before any PHI-shaped (synthetic) data flows through those integrations during testing.

18.4.1 Dependencies map

Dependency	Type	Owner	Needed By	Status	Mitigation of Late
CI/CD pipeline (SAST/SCA/Container scan)	Upstream	Team A	M2	Yellow	Local pre-commit hooks + manual nightly scans by QA Lead.
AWS Bedrock / Claude access	External	AWS / Instructor	M2	Yellow	Use mock Bedrock client and recorded fixtures.
EVV aggregator endpoint	External	State Aggregator	M3	Red	Schema-only contract tests + EDI 837 synthetic batch; deliver REQ-1404 documentation.
Wearable sandboxes (Fitbit/HealthKit/Health Connect)	External	Vendors	M2	Yellow	OAuth-layer mocks; record-and-replay HTTP fixtures.
Approved Figure 2 ERD (Tech Design 7.2-7.6)	Documentation	Backend Lead	M2	Green	Schema-first test stubs; freeze schema at end of Sprint 3.
Backend API contracts (per sprint)	Internal	Backend Lead	End of every sprint	Green	Contract testing via Pact; OpenAPI diff alerts.
BAA verification	External	Instructor / UMGC	M2	Green	Restrict synthetic data flow to BAA-covered services only.
Frontend build artifacts for E2E	Internal	Frontend Lead	End of every sprint	Green	Defer E2E to the following sprint; extend API-level coverage.

19. Approval and Sign-Off

Name	Role	Review / Approval Status	Date
Davon Davis	Team Lead	Pending Review	TBD
Bernabe Garcia	Test Owner	Pending Review	TBD
Dr. Assadullah	Instructor	Pending Review	TBD
Roy Gordon	Client/Stakeholder	Pending Review	TBD

20. Appendices

Appendix A: Test Case Inventory

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-EVV-001	Valid GPS check-in within geofence transitions visit to IN_PROGRESS	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-002	Out-of-geofence check-in is rejected with validation message	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-003	Valid check-out transitions visit from IN_PROGRESS to COMPLETED	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-004	Check-out required fields are validated before completion	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-005	Visit correction captures before and after snapshot	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-006	Unauthorized role cannot submit EVV correction	Security/RBAC	High	Bernabe Garcia	Not Run
TC-D-EVV-007	Pre-export validation blocks missing required aggregator fields	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-008	Complete EVV record passes pre-export validation	Functional/API	High	Cristobal Garcia	Not Run
TC-D-EVV-009	ARRIVE audit entry is written during check-in	Compliance/Audit	High	Bernabe Garcia	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-EVV-010	DEPART audit entry is written during check-out	Compliance/Audit	High	Bernabe Garcia	Not Run
TC-D-AGG-001	Aggregator dashboard displays PENDING, SUBMITTED, and FAILED statuses	UI/Integration	High	Cristobal Garcia	Not Run
TC-D-AGG-002	Aggregator dashboard supports 10 or more configured aggregator records	UI/Integration	High	Cristobal Garcia	Not Run
TC-D-AGG-003	Aggregator status card displays last updated timestamp	UI	Medium	Cristobal Garcia	Not Run
TC-D-AGG-004	EVV external submission remains mock or preview-only when live target is unavailable	Reliability/Integration	High	Cristobal Garcia	Not Run
TC-D-INV-001	Invoice upload accepts allowed file types and size limit	Functional/API	High	Cristobal Garcia	Not Run
TC-D-INV-002	Invoice upload rejects unsupported file type or oversized file	Negative/API	High	Cristobal Garcia	Not Run
TC-D-INV-003	Invoice attachment is stored with SSE-KMS encryption	Security/Integration	High	Cristobal Garcia	Not Run
TC-D-INV-004	Mock Bedrock extraction returns required fields and confidence scores	Functional/Integration	High	Cristobal Garcia	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-INV-005	Bedrock failure routes invoice to manual review with controlled error	Reliability/Integration	High	Cristobal Garcia	Not Run
TC-D-INV-006	Reviewer field correction creates invoice audit entry	Compliance/Audit	High	Cristobal Garcia	Not Run
TC-D-INV-007	Reviewer continuity is preserved during invoice re-review	Functional	High	Cristobal Garcia	Not Run
TC-D-INV-008	Invoice cannot be approved without explicit human approval	Compliance	Critical	Cristobal Garcia	Not Run
TC-D-INV-009	Duplicate invoice detection requires reviewer acknowledgment	Functional	Medium	Cristobal Garcia	Not Run
TC-D-INV-010	Automated high-confidence invoice cannot bypass approval gate	Compliance/Security	Critical	Bernabe Garcia	Not Run
TC-D-INVITE-001	Invite token is generated with HMAC-SHA256 signature	Functional/Security	High	Nhial Mayar	Not Run
TC-D-INVITE-002	Raw invite token is not stored; only SHA-256 hash persists	Security	High	Nhial Mayar	Not Run
TC-D-INVITE-003	Invite is delivered as QR code and copyable URL	UI/Functional	High	Nhial Mayar	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-INVITE-004	Valid token returns invite metadata and correct onboarding path	Functional	High	Nhial Mayar	Not Run
TC-D-INVITE-005	Expired token returns controlled expired-token response	Negative/Security	High	Nhial Mayar	Not Run
TC-D-INVITE-006	Already-used token returns controlled replay response	Negative/Security	High	Nhial Mayar	Not Run
TC-D-INVITE-007	Revoked invite token cannot be accepted	Security	High	Nhial Mayar	Not Run
TC-D-INVITE-008	Accepted invite assigns correct user role	Functional/RBAC	High	Nhial Mayar	Not Run
TC-D-WEAR-001	Health Connect permission grant creates wearable connection	Functional/Integration	High	Davon Davis	Not Run
TC-D-WEAR-002	Permission denial displays user guidance	UI/Negative	High	Davon Davis	Not Run
TC-D-WEAR-003	Batch metric sync persists valid wearable records	Functional/Integration	High	Davon Davis	Not Run
TC-D-WEAR-004	Dashboard displays latest wearable metric per metric type	UI/Functional	High	Davon Davis	Not Run
TC-D-WEAR-005	Threshold breach dispatches caregiver alert	Functional/Integration	High	Davon Davis	Not Run
TC-D-WEAR-006	Threshold alert cooldown prevents repeated alert flooding	Functional/Reliability	High	Davon Davis	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-WEAR-007	Consent revocation stops future wearable syncs	Security/Privacy	High	Davon Davis	Not Run
TC-D-WEAR-008	Batch of 100 wearable metrics processes within target latency	Performance	Medium	Davon Davis	Not Run
TC-D-DOC-001	Document upload creates record in UPLOADED state	Functional/API	High	Bernabe Garcia	Not Run
TC-D-DOC-002	W-4, I-9, and Direct Deposit document types are accepted	Functional/API	High	Bernabe Garcia	Not Run
TC-D-DOC-003	Document storage uses SSE-KMS encryption	Security/Integration	High	Bernabe Garcia	Not Run
TC-D-DOC-004	Invalid document file type or missing metadata is rejected	Negative/API	Medium	Bernabe Garcia	Not Run
TC-D-DOC-005	Family Read-Only role cannot submit document endpoint	Security/RBAC	High	Bernabe Garcia	Not Run
TC-D-DOC-006	Document review status transitions from submitted to accepted or returned	UI/Functional	Medium	Bernabe Garcia	Not Run
TC-D-SCREEN-001	Caregiver creates screening record in DRAFT state	Functional/API	High	Angel Guanlao	Not Run
TC-D-SCREEN-002	Screening draft saves on field update	Functional	High	Angel Guanlao	Not Run
TC-D-SCREEN-003	Required screening fields are validated before submission	Functional/Negative	High	Angel Guanlao	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-SCREEN-004	Submitted screening record is immutable	Compliance/Functional	High	Angel Guanlao	Not Run
TC-D-SCREEN-005	Patient role is denied access to screening submission API	Security/RBAC	High	Angel Guanlao	Not Run
TC-D-ASSESS-001	Comprehensive assessment record is created with five sections	Functional/API	High	Angel Guanlao	Not Run
TC-D-ASSESS-002	Assessment supports section-level draft save	Functional	High	Angel Guanlao	Not Run
TC-D-ASSESS-003	Assessment submission requires all five sections	Functional/Negative	High	Angel Guanlao	Not Run
TC-D-ASSESS-004	Submitted assessment is immutable	Compliance/Functional	High	Angel Guanlao	Not Run
TC-D-ASSESS-005	Assessment amendment creates linked new record without changing original	Functional/Compliance	High	Angel Guanlao	Not Run
TC-D-ASSESS-006	Assessment retains historical versions for longitudinal review	Functional	Medium	Angel Guanlao	Not Run
TC-D-SEC-001	RBAC enforcement returns 403 for unauthorized Team D endpoints	Security/RBAC	High	Bernabe Garcia	Not Run
TC-D-SEC-002	PHI and sensitive values are excluded from logs	Security/Privacy	Critical	Bernabe Garcia	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-SEC-003	TLS 1.3 is enforced for Team D API traffic	Security/NFR	High	Bernabe Garcia	Not Run
TC-D-SEC-004	Direct UPDATE on audit_event is rejected by database trigger	Security/Compliance	Critical	Bernabe Garcia	Not Run
TC-D-SEC-005	Direct DELETE on model_prompt_log is rejected by database trigger	Security/Compliance	Critical	Bernabe Garcia	Not Run
TC-D-SEC-006	Audit hash-chain verification detects tampered audit row	Security/Compliance	Critical	Bernabe Garcia	Not Run
TC-D-SEC-007	AI prompt redaction prevents PHI submission to model provider	Security/Privacy	Critical	Bernabe Garcia	Not Run
TC-D-SEC-008	Invite token database record stores hash only and no raw token	Security	High	Bernabe Garcia	Not Run
TC-D-PERF-001	EVV check-in and check-out response time meets P95 target	Performance	Medium	Cristobal Garcia	Not Run
TC-D-PERF-002	Invite token validation latency meets P95 target	Performance	Medium	Nhial Mayar	Not Run
TC-D-PERF-003	Wearable batch sync processes within target time	Performance	Medium	Davon Davis	Not Run
TC-D-PERF-004	Assessment draft save remains responsive with realistic form size	Performance	Medium	Angel Guanlao	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-REL-001	AI provider unavailable fails safely and preserves review context	Reliability	High	Cristobal Garcia	Not Run
TC-D-REL-002	Repeated invoice corrections do not duplicate approvals or lose edits	Reliability	High	Cristobal Garcia	Not Run
TC-D-REL-003	Interrupted assessment draft can be resumed without data loss	Reliability	High	Angel Guanlao	Not Run
TC-D-REL-004	Malformed wearable payload is rejected without data corruption	Reliability/Negative	Medium	Davon Davis	Not Run
TC-D-BAT-001	System health endpoint returns HTTP 200	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-D-BAT-002	Authentication succeeds and MFA prompt triggers	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-D-BAT-003	Caregiver token attempting admin route receives 403	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-D-BAT-004	EVV check-in and check-out happy path completes	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-D-BAT-005	Invoice upload succeeds and mock Bedrock callback returns 200	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-D-BAT-006	Synthetic wearable data point posts successfully to telemetry endpoint	Smoke/BAT	Medium	Bernabe Garcia	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-D-BAT-007	Audit event row is written for login action	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-D-BAT-008	Audit event UPDATE attempt is rejected by trigger	Smoke/BAT	Critical	Bernabe Garcia	Not Run
TC-D-BAT-009	API gateway rejects deprecated TLS versions	Smoke/BAT	High	Bernabe Garcia	Not Run
TC-TD-UI-001	EVV dashboard lifecycle state visibility	UI	High	Team D QA/Frontend	Not Run
TC-TD-UI-002	EVV record correction and retry interaction	UI	High	Team D QA/Frontend	Not Run
TC-TD-UI-003	Invoice upload and extraction preview display	UI	High	Team D QA/Frontend	Not Run
TC-TD-UI-004	Invoice reviewer field correction and approval UI	UI	High	Team D QA/Frontend	Not Run
TC-TD-UI-005	Onboarding token intake and validation state messaging	UI	High	Team D QA/Frontend	Not Run
TC-TD-UI-006	Role routing after successful invite onboarding	UI	Medium	Team D QA/Frontend	Not Run
TC-TD-UI-007	Wearable connection status and sync freshness display	UI	Medium	Team D QA/Frontend	Not Run
TC-TD-UI-008	Wearable alert display and acknowledgment flow	UI	Medium	Team D QA/Frontend	Not Run
TC-TD-UI-009	Home-care form draft/save/submit interactions	UI	High	Team D QA/Frontend	Not Run

Test Case ID	Title	Type	Priority	Owner	Status
TC-TD-UI-010	Document review status transitions	UI	Medium	Team D QA/Frontend	Not Run
TC-TD-UI-011	Screening form required field validation behavior	UI	High	Team D QA/Frontend	Not Run
TC-TD-UI-012	Comprehensive assessment multi-section progression	UI	High	Team D QA/Frontend	Not Run
TC-TD-REG-001	EVV lifecycle regression	Regression	High	Team D QA	Not Run
TC-TD-REG-002	EVV retry path regression	Regression	High	Team D QA	Not Run
TC-TD-REG-003	Invoice review workflow regression	Regression	High	Team D QA	Not Run
TC-TD-REG-004	Onboarding token handling regression	Regression	High	Team D QA	Not Run
TC-TD-REG-005	Wearable status and alert regression	Regression	Medium	Team D QA	Not Run
TC-TD-REG-006	Document lifecycle regression	Regression	High	Team D QA	Not Run
TC-TD-REG-007	Assessment draft and submit regression	Regression	High	Team D QA	Not Run

For this milestone draft, Appendix A is intentionally focused on Team D UI and regression tests only, as requested. Cross-team or non-UI expansions can be appended later.

Appendix B: Test Execution Log

Date	Test Case ID	Tester	Result	Evidence Location	Notes
2026-06-07	NA	TEAM D	Not Run	NA	Milestone planning phase. Execution evidence pending scheduled test runs.

Appendix C: Defect Log

ID	Title	Module	Linked REQ ID	ERD Entity	Severity	Status	Compliance Impact
DEF-001	Offline sync queue fails to retry after 24h expiry	EVV	REQ-6.4	offline_sync_queue	Sev-2	In-Progress	Y — Cures Act §12006
DEF-002	Human approval gate bypassed on high-confidence invoice	Invoicing	REQ-1507	invoice	Sev-1	Fixed	Y — Internal Policy + REQ-1507
DEF-003	UPDATE command succeeds on audit log	Audit	HIPAA §164.312(b)	audit_event	Sev-1	Verified	Y — HIPAA §164.312(b)
DEF-004	Wearable threshold alert not delivered to caregiver for HR > 130	Wearable	REQ-093	wearable_data_point	Sev-2	New	N
DEF-005	Ask AI response missing required medical-judgment disclaimer	AI	REQ-5.7.2-002	model_prompt_log	Sev-2	Triaged	Y — Internal Compliance Policy

ID	Title	Module	Linked REQ ID	ERD Entity	Severity	Status	Compliance Impact
DEF-006	Encryption-at-rest verification fails for invoice S3 bucket (missing SSE-KMS)	Identity/Audit	REQ-6.1	invoice , S3 attachment bucket	Sev-1	In-Progress	Y — HIPAA §164.312(a)(2)(iv)
DEF-007	Family Read-Only user successfully submitted a calendar update	Family / Social	REQ-5.11.2-002	user , role , permission	Sev-1	New	Y — HIPAA §164.312(a)(1)

Appendix D: AI Coordination Summary Template

Date	Activity Type	Tool Used	Prompt Summary (NO PHI)	Output Summary	Reviewer	Accepted (Y/N)	Modifications	Linked Artifact	ModelPromptLog ID
06/02/2026	Test Generation	Claude 3.5	Generate edge-case tests for EVV Geofence (REQ-1402)	5 synthetic boundary scenarios	Bernabe G.	Y	Adjusted GPS radii limits	TC-EVV-054	UUID-8a7b...

Date	Activity Type	Tool Used	Prompt Summary (NO PHI)	Output Summary	Reviewer	Accepted (Y/N)	Modifications	Linked Artifact	ModelPromptLog ID
06/03/2026	Data Synth	ChatGPT 4o	Generate 50 synthetic GPS traces near/around 100m geofence boundary	CSV with 50 trace records	Bernabe G.	Y	Trimmed to 20 high-signal traces	fixtures/geo_traces.csv	UUID-2d3e...
06/04/2026	Code Review	GitHub Copilot	Review newly authored RBAC enforcement tests for missing role coverage	Identified 2 missing role permutations (Invited Pending, Family Read-Only)	Bernabe G.	Y	Added 2 test cases	PR #142	UUID-5f6a...
06/05/2026	Data Synth	ChatGPT 4o	Create 10 fake invoice JSONs matching DB schema	10 JSON payloads	Nhial M.	Y	None	Commit 9f2a3c	UUID-4f1c...

Date	Activity Type	Tool Used	Prompt Summary (NO PHI)	Output Summary	Reviewer	Accepted (Y/N)	Modifications	Linked Artifact	ModelPromptLog ID
06/06/2026	Defect Triage Assist	Claude 3.5	Suggest probable root cause categories for DEF-004 (wearable alert miss)	Three hypotheses with diagnostic queries	Bernabe G.	Partial	Used hypothesis #2; discarded #1 and #3	DEF-004	UUID-9b1d...